

Chelsea Chukwudozie

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PROFILE

I am a final-year Computer Science student at Nottingham Trent University with a strong interest in intelligent systems, machine learning, and scalable software development. I have experience developing predictive models, conversational AI systems, and full-stack analytical applications using Python, Scikit-learn, TensorFlow, Flask, and RESTful APIs. My work has involved Ensemble Learning, CNNs, NLP, Symbolic Reasoning, and explainable AI, particularly within FemTech and data-driven research projects. I enjoy building structured, user-centred systems that combine technical accuracy with practical usability.

TECHNICAL SKILLS

Languages: Python, SQL, HTML/CSS, JavaScript

Machine Learning: Scikit-learn, TensorFlow, CNNs, Ensemble Learning, NLP

Frameworks & Tools: Flask, Docker, REST APIs, Pandas, NumPy, Git

Concepts: CRISP-DM, Agile Development, Explainable AI, Predictive Modelling

EDUCATION

Nottingham Trent University

BSc Computer Science | 2023 – Present

On track for First Class

PROFESSIONAL MEMBERSHIPS

Royal Society for Statistics | Association for Project Management

FINAL YEAR PROJECT

HerSignal

Intelligent Conversational System for PCOS Awareness and Pattern Recognition

Final Year Project | Yet to be Graded

- Designed and developed a multi-layer conversational system for PCOS symptom awareness using a hybrid AI and rule-based architecture.
- Developed a conversational pipeline focused on explainability, controlled retrieval, and response safety.
- Built a dense retrieval FAQ system using sentence-transformers, cosine similarity, fuzzy matching, and NLP-based ranking techniques.
- Designed a JSON-configurable scoring model for hormonal, metabolic, and inflammatory symptom analysis.
- Conducted user-centred research to refine functionality, interaction design, and educational support features.
- Optimised latency using LRU embedding caching and implemented secure session handling and FAQ rate-limiting protections.

- Developed a responsive Flask-based application with chart visualisation, PDF export, onboarding flows, and insight tracking.

SELECTED ACADEMIC PROJECTS

BloomBud

Intelligent Multi-Modal AI Chatbot | Upper First

- Engineered a modular multi-modal system integrating Computer Vision (CNN), Symbolic Reasoning, and NLP via a RESTful architecture.
- Developed a hybrid inference layer using rule-based logical structures and TF-IDF similarity matching over AIML.
- Built and integrated a TensorFlow/Keras CNN for real-time botanical classification with confidence-based fallbacks.
- Containerised components using Docker and implemented cloud integrations to support scalability and dynamic data retrieval.

FFsmart

Smart Inventory & Food Management System | Project Lead | High 2:1

- Led the analysis, interface design, and prototype development of a collaborative inventory management platform.
- Coordinated development tasks within a cross-functional Agile team throughout the SDLC.
- Translated stakeholder requirements into workflows, use cases, and Balsamiq wireframes for implementation planning.
- Developed modular UI components for inventory tracking, expiry alerts, and workflow management.

ML Predictive System

Developer Income Classification | First Class

- Built an end-to-end ML pipeline using the CRISP-DM framework to classify high-income developers from a 65,000+ record dataset.
- Developed preprocessing workflows involving feature selection, encoding, normalisation, and statistical outlier detection.
- Benchmarked supervised learning models including Logistic Regression, Decision Trees, and k-NN classifiers.
- Improved predictive performance through hyperparameter tuning and ensemble learning techniques.